antibodies using serum from humans or animals. This type of immunity is useful for providing immediate protection against acute infections like tetanus, measles etc.

2. Active Immunization

Active immunization is the administration of vaccines containing microbial products with or without adjuvants in order to obtain long term immunological protection against the offending microbe.

At present the normal route of vaccination in most instances is either intramuscularly or subcutaneously.

Oral immunization is the method of choice for polio and *Salmonella typhi* vaccines. However, there is an increasing awareness that this route of immunization may be the best for most immunizations since nearly all infectious agents gain entrance through the mucosal surfaces.

Active natural immunization involves activation of immune system in the body to produce antibodies. It is achieved in both clinical and subclinical infections.

Active artificial immunization is achieved by the administration of vaccines or toxoids.

Antigen preparations

Most vaccines consist of attenuated organisms, killed organisms, inactivated toxins, or sub cellular fragments and more recently genes for antigens in viral 'vectors', and DNA itself. Thus, vaccines must be capable of targeting the immune system appropriately i.e. cellular/or humoral mechanisms (Table 11.3).

Table 11.3: Antigen Preparations Used in Vaccines

Type of antigen	Examples	
	Viruses	**Bacteria**
Normal heterologous organism	Vaccinia (Cowpox)	
Living attenuated organism	Measles	BCG
	Mumps	Typhoid (New)
	Rubella	
	Polio (Sabin)	
	Yellow fever	
	Varicella-Zoster	
Whole killed oranism	Rabies	Pertussis
	Poli (Salk)	Typhoid
	Influenza	Cholera
Inactivated toxin (toxoid)		Diphtheria
		Tetanus
		Cholera (New)
Capsular polysaccharide		*Meningococcus*
		Pneumococcus
		Haemophilus
		Typhoid (New)
Surface antigen	Hepatitis B	

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